

Submission: Week 08 Hardening & Quality Audit

Break Your Own Site • SEO, Performance & Hardening Review Package

Why It Matters:

"Anyone can demo the happy path. The professional difference is knowing exactly where your thing breaks, the empty input, the weird browser, the search that finds nothing. Finding your own cracks and being honest about them is the Diligence skill employers actually trust."

- Live Portfolio URL: <https://alexchen-ai.netlify.app>
- Hardening Review Partner: Peer Reviewer (Structured Track Review) / Mentor Approved

1. Edge-Case Testing & "Where It Breaks" Triage

To test system limits beyond the happy path, rigorous edge-case testing was performed across multiple browsers, network conditions, and input variations. All findings were triaged into 'Fix-Now' (resolved before submission) and 'Known Limitations' (documented).

Test Category / Edge Case	Observed Failure / Behavior	Triage Status	Resolution / Mitigation Action
Empty Form & Garbage Input	Form allowed whitespace-only strings and unformatted emails.	Fix-Now	Added HTML5 client-side validation (required, regex pattern) & backend sanitization.
Rapid Double Submission	Submitting twice quickly triggered duplicate emails and API calls.	Fix-Now	Disabled submit button immediately on click and added a loading spinner state.
Cross-Browser & Safari Mobile	Layout shifted on Safari Mobile due to unsupported height: 100vh on iOS.	Fix-Now	Replaced 100vh with dynamic 100dvh CSS viewport units for mobile browser compatibility.
Search / Filter Finds Nothing	UI showed blank screen when portfolio tag filter returned zero items.	Fix-Now	Implemented explicit 'No matching projects found. Reset filters' empty state message.

Offline / Spotty Network Connection	Form submission hung indefinitely with no user feedback when network failed.	Known Limitation	Logged network error to console. Full offline offline-first PWA caching deferred to v2.
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2. SEO, Meta Tags & Speed Benchmark Audit

Basic SEO metadata and Open Graph social sharing tags were added to the `<head>` of `index.html` to ensure findability and proper link previews across platforms.

Added HTML Meta Tags Snippet:

```
``html
<title>Alex Chen | AI Engineer & Full-Stack Portfolio</title>
<meta name="description" content="Portfolio of Alex Chen showcasing end-to-end AI agent systems and web integrations.">
<meta property="og:title" content="Alex Chen - AI Systems Portfolio">
<meta property="og:description" content="Explore live working demos and production AI projects.">
<meta property="og:image" content="https://alexchen-ai.netlify.app/assets/og-preview.png">
<meta property="og:type" content="website">
...

```

PageSpeed Insights Performance Audit Results:

Performance Category	PageSpeed Score	Audit Status & Notes
Performance	98 / 100	Fast First Contentful Paint (0.8s); static assets cached.
Accessibility	100 / 100	High-contrast colors and full keyboard navigation supported.
Best Practices	100 / 100	HTTPS enforced, zero browser console errors.
SEO	100 / 100	Meta description, page title, structured markup, and canonical URL present.

3. Peer/Mentor Hardening Review Summary

A structured hardening review was conducted with a peer track reviewer. The reviewer was tasked with attempting to break the site and reviewing the triage list.

Reviewer Feedback & Must-Fix Directives:

1. Feedback: "The double-click prevention works, but there is no visual indicator while sending. Users might think it froze."

- **Fix Applied:** Added an inline CSS spinner and changed button text to 'Sending...' immediately upon form submission.

2. Feedback: "Social share card (og:image) URL was returning a 404 image link."

- **Fix Applied:** Updated metadata image path to absolute production URL and verified rendering via Twitter/LinkedIn card debuggers.

4. Checkpoint Pass / Revise Criteria Self-Audit

Evaluation Criteria	Status	Verification Details
Genuinely tried to break site with real edge cases	[PASS]	Tested empty input, double clicks, cross-browser Safari iOS, and bad queries.
Basic SEO/meta added; site findable & speed checked	[PASS]	Open Graph tags verified; PageSpeed Insights score: 98/100.
Findings triaged honestly; fix-nows fixed & limitations named	[PASS]	4 fix-nows resolved; 1 offline limitation clearly documented.
Hardening review performed & must-fixes addressed	[PASS]	Peer review conducted; loading spinner state & image URL fixed.